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Introduction

An odd prime number p is called *regular* if p does not divide the class number of $\mathbf{Q}(\mu_p)$. Otherwise it is called *irregular*.

In 1857, Kummer proved the FLT for regular primes. He also provided a criterion for when a prime is regular in terms of Bernoulli numbers.

The k-th *Bernoulli number* B_k is given by

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} \frac{B_n}{n!} t^n.$$

The first few Bernoulli numbers are

$$B_0 = 1, \quad B_2 = \frac{1}{6}, \quad B_4 = \frac{-1}{30}, \quad B_6 = \frac{1}{42}, \quad B_8 = \frac{-1}{30}, \quad B_{10} = \frac{5}{66}, \quad B_{12} = \frac{-691}{2730}.$$

All the odd Bernoulli numbers are all zero except for $B_1 = -1/2$.

(Kummer)**.** *An odd prime is irregular iff p divides the numerator of the B_k, for an even integer $2 \leq k \leq p-2$.*

Thus, we see that 691 is an irregular prime, while 3, 5, 7, 11, 13 are regular. In fact, it is easy to prove that there are infinitely many irregular primes. However, it is not known today whether there are infinitely many regular primes. Siegel conjectured (1964) that $e^{-0.5}$ (about 61%) of all prime numbers are regular, in the asymptotic sense.

What is significant about Bernoulli numbers is that they are *analytic* objects, because they come from special values of L-functions:

$$\zeta(1 - n) = \frac{-B_n}{n}$$

for $n \geq 1$. (Note the trivial zeros.) In fact, one can define generalized Bernoulli numbers $B_{n,\chi}$ with respect to a Dirichlet character χ so that one has $L(1 - n, \chi) = -B_{n,\chi}/n$.

A great deal of advances in number theory have to do with connecting special values of L-functions to arithmetic phenomena. One example of this is the theorem of Herbrand, which shows that p divides a Bernoulli number if a particular eigenspace of a natural Galois action on the p-part of the ideal class group of $\mathbf{Q}(\mu_p)$ is nonzero.

Let A be the class group of $\mathbf{Q}(\mu_p)$ and let C be the $\mathbf{F}_p$-vector space A/A^p. Note that $\dim_{\mathbf{F}_p} C$ is the p-rank of A. The absolute Galois group acts on the $\mathbf{F}_p$-vector space C through its abelian quotient $\mathrm{Gal}(\mathbf{Q}(\mu_p)/\mathbf{Q}) \simeq (\mathbf{Z}/p\mathbf{Z})^\times$. By Maschke's theorem, since $|(\mathbf{Z}/p\mathbf{Z})^\times| = p-1$ is prime to $p = \mathrm{char}(\mathbf{F}_p)$, this representation is semisimple, so it decomposes

$$C = \bigoplus_{i \mod p-1} C(\chi^i), \quad C(\chi^i) = \{c \in C \mid \sigma(c) = \chi^i(\sigma) \cdot c\}$$

where $\chi : \mathrm{Gal}(\mathbf{Q}(\mu_p)/\mathbf{Q}) \to \mathbf{F}_p^\times$ is the standard cyclotomic character defined by $\sigma(\mu_p) = \mu_p^{\chi(\sigma)}$ and $C(\chi^i)$ is the χ^i-isotypical component of C as a Galois module.

(Herbrand, 1932)**.** *Let $2 \leq k \leq p-3$ be an even integer. If $C(\chi^{1-k}) \neq 0$, then p divides the numerator of B_k.*

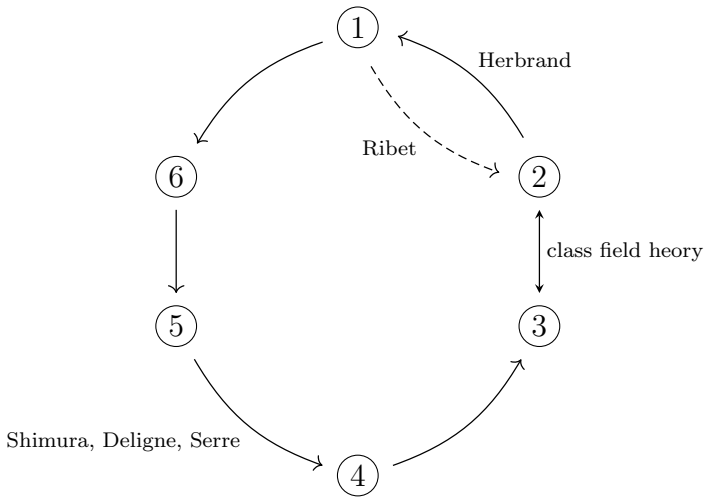
Herbrand's proof is classical. The converse implication, which is proved by Ribet in his 1976 paper *A modular construction of unramified p-extensions of $\mathbf{Q}(\mu_p)$*, will be our focus today.

Theorem (1.2 of Ribet)**.** *Let $2 \leq k \leq p-3$ be an even integer. If p divides the numerator of B_k, then $C(\chi^{1-k}) \neq 0$.*

The Herbrand-Ribet describes the natural Galois action on the p-part of the class group of $\mathbf{Q}(\mu_p)$ in terms of analytic quantities. However, this is not a complete description. We will briefly discuss some further descriptions towards the end of the talk. In fact, Mazur-Wiles' later proof of the main conjecture of Iwasawa theory 8 years later in 1984 implies Ribet's result as a corollary, but Ribet's idea and technique of explicitly constructing an abelian unramified p-extension of $\mathbf{Q}(\mu_p)$ with controlled behavior remains valuable and indeed inspired many later results.

Fun fact: Ribet's paper is awarded the Steele Prize for Seminal Contribution to Research in 2025.

Let me briefly summarize Ribet's overall strategy using the following heuristic illustration taken from Mazur's article.



Here, (1) is Bernoulli numbers, (2) is ideal class groups, (3) is abelian extensions of cyclotomic fields, (4) is Galois representations, (5) is cusp eigenforms, and (6) is Eisenstein series. Note that the objects on the left hand side are modular forms, and the right hand side belongs to algebraic number theory. Bernoulli numbers (analytic) and Galois representations (representation-theoretic) are bridges between these two realms.

To prove the desired result, Ribet

(i) start from properties of Bernoulli numbers B_k to

(ii) note a consequence about Eisenstein series, (namely, their constant terms are given by Bernoulli numbers), which

(iii) allows him to construct certain cusp Hecke eigenforms that are congruent to these Eisenstein series modulo p, and in turn to

(iv) construct, using these cuspforms, a certain 2-dimensional $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$-representation that

(v) cuts out a number field $L \subset \overline{\mathbf{Q}}$ consisting of all the algebraic numbers fixed by the kernel of the Galois representation;

(vi) and finally, an easy exercise from class field theory implies that the existence of such number field guarantees the nontriviality of $C(\chi^{1-k})$.

(1) $\Rightarrow$ (6)

I won't go into details about modular forms due to time constraints. I will nevertheless define them for completeness. I apologize if this rash development of the theory seems rather unmotivated. A *congruence subgroup* of $\mathrm{SL}_2(\mathbf{Z})$ is a subgroup that contains

$$\Gamma(N) = \left\{ \gamma \in \mathrm{SL}_2(\mathbf{Z}) : \gamma \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mod N \right\}$$

for some positive integer N. Two important examples of congruence subgroups are

$$\Gamma_1(N) = \left\{ \gamma \in \mathrm{SL}_2(\mathbf{Z}) : \gamma \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \mod N \right\}, \quad \text{and}$$
$$\Gamma_0(N) = \left\{ \gamma \in \mathrm{SL}_2(\mathbf{Z}) : \gamma \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \mod N \right\}$$

Obviously, we have $\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N) \subset \mathrm{SL}_2(\mathbf{Z})$. A *modular form* of weight k (an integer) with respect to a congruence subgroup $\Gamma(N) \subset \Gamma \subset \mathrm{SL}_2(\mathbf{Z})$ is a holomorphic function on the upper half plane $f : \mathcal{H} = \{z \in \mathbf{C} : \mathrm{Im}(z) > 0\} \to \mathbf{C}$ that satisfies

$$f(\gamma(z)) = (cz + d)^k f(z), \quad \text{for} \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma.$$

Actually, there is another requirement for f to be a modular form (called holomorphic at cusps): if I express f as a function of $q_N = ^{2\pi iz/N}$:

$$f = \sum_{n \geq 0} a_n q_N^n$$

(this is called the q-expansion or Fourier expansion), then this Fourier expansion extends to a holomorphic function at $q = 0$, and $|a_n| \leq Cn^r$ for all $n > 0$ for some constants $C, r > 0$. If the constant term of the Fourier expansion of $(ck + d)^{-k} f(\gamma(z))$ is 0 for all $\gamma \in \mathrm{SL}_2(\mathbf{Z})$, then f is called a *cusp form*.

Ribet worked with cusp forms of weight $k = 2$, so we will supply another equivalent, perhaps more natural (to those algebraic-minded) definition. Note that a congruence subgroup Γ acts on the upper half plane $\mathcal{H}$ via fractional linear transforms. The *modular curve* $Y(\Gamma)$ is defined to be the quotient space $Y(\Gamma) = \Gamma \setminus \mathcal{H}$ of this action, and $X(\Gamma) = \Gamma \setminus (\mathcal{H} \cup \{\infty\} \cup \mathbf{Q})$ is the compact version of $Y(\Gamma)$. The modular curves $X(\Gamma)$ and $Y(\Gamma)$ are Riemann surfaces. Then, the space of cusp forms of weight $k = 2$, denoted $S_2(\Gamma)$, is identified with the $\mathbf{C}$-vector space of holomorphic differential 1-forms on $X(\Gamma)$, via the map $f \mapsto f(dz)$.

Finally, we also need the notion of *nebentypus*. The idea is that since $\Gamma_1(N) \subset \Gamma_0(N)$, a modular form w.r.t $\Gamma_1(N)$ is not necessarily modular w.r.t. $\Gamma_0(N)$, but it is nearly so up to a twist. For a Dirichlet character $\varepsilon : (\mathbf{Z}/N\mathbf{Z})^\times \to \mathbf{C}^\times$, a modular form $f \in M_k(\Gamma_1(N))$ is of *type* ε if

$$f(\gamma(z)) = \varepsilon(d)(cz + d)^k f(z), \quad \text{for} \quad \gamma \in \Gamma_0(N).$$

For $N > 1$, there is a decomposition $M_k(\Gamma_1(N)) = \oplus_\varepsilon M_k(N, \varepsilon)$ according to the type.

Examples of modular forms that are particular to our interest are the *Eisenstein series*:

$$G_k(z) = -\frac{B_k}{2k} + \sum_{n \geq 1} \sigma^{k-1}(n) q^n, \quad \text{where} \quad \sigma^{k-1}(n) = \sum_{d|n} d^{k-1}, \quad k \geq 4 \text{ even}$$

are modular forms of weight k with respect to $\mathrm{SL}_2(\mathbf{Z})$. Fixing a nontrivial even type ε (i.e., $\varepsilon(-1) = 1$) on $(\mathbf{Z}/p\mathbf{Z})^\times$, we also define an Eisenstein series

$$G_{2,\varepsilon} = \frac{L(-1, \varepsilon)}{2} + \sum_{n \geq 1} \sum_{d|n} \varepsilon(d) d q^n,$$

which is a modular form of weight 2. Fixing an odd type ε on $(\mathbf{Z}/p\mathbf{Z})^\times$, the Eisenstein series of weight 1 is defined by

$$G_{1,\varepsilon} = \frac{L(0, \varepsilon)}{2} + \sum_{n \geq 1} \sum_{d|n} \varepsilon(d) q^n.$$

(6) $\Rightarrow$ (5)

If p divides the numerator of B_k (we shall simply write $p \mid B_k$), then $G_k(z)$ ought to "look like" a cusp form modulo p. That is precisely what we are trying to construct here. But that is not enough: we need these cusp forms to be *Hecke eigenforms*.

There is an operator $\langle d \rangle$, commonly called the *diamond operator*, that acts on $M_k(\Gamma_1(N))$ such that if $f \in M_k(N, \varepsilon)$, then $\langle d \rangle f = \varepsilon(d) f$ for all $d \in (\mathbf{Z}/N\mathbf{Z})^\times$. If $(d, N) > 1$, then define $\langle d \rangle = 0$. The action restricts to cusp forms and preserve types.

There is another operator T_p, called a *Hecke operator*, that acts on $M_k(\Gamma_1(N))$ as follows. Let $f(z) = \sum_{n \geq 0} a_n(f) q^n \in M_k(N, \varepsilon)$. Then

$$T_p f(z) = \sum_{n \geq 0} \left(a_{np}(f) + \varepsilon(p) p^{k-1} a_{n/p}(f) \right) \right) q^n, \quad \text{where} \quad a_{n/p}(f) = 0 \quad \text{if} \quad n/p \notin \mathbf{Z}.$$

The action again restricts to $S_k(\Gamma_1(N))$ and $M_k(N, \varepsilon)$. In fact, T_n can be defined for any $n > 0$ in terms of the T_p's via $T_m T_n = T_{mn}$ for $(m, n) = 1$, and $T_{p^r} = T_p T_{p^{r-1}} - p^{r-1} \langle p \rangle T_{p^{r-2}}$ for $r \geq 2$.

The Hecke operators $\langle n \rangle$ and T_n are pairwise commute for $(n, N) = 1$. We can define a *Hecke algebra* $\mathbf{T}^0$ to be the $\mathbf{Z}$-algebra generated by $\langle n \rangle$ and T_n for $(n, N) = 1$, that acts on $S_k(\Gamma_1(N))$. One can define a inner product on the finite-dimensional vector space $S_k(\Gamma_1(N))$ (called the Petersson inner product) such that $\langle n \rangle$ and T_n are normal (commute with their adjoint operators). Hence, the spectral theorem from linear algebra implies that $S_k(\Gamma_1(N))$ has an orthogonal basis of simultaneous eigenforms for the $\mathbf{T}^0$.

(Note: each level has its own Hecke algebra, but N is usually omitted from the notation.)

In fact, the theory of newforms due to Atkin and Lehner provides criterion when a cusp form can be a simultaneous eigenform for the full Hecke algebra $\mathbf{T}$ (dropping the condition $(d, N) = 1$). In our case, we are primarily interested in $S_2(\Gamma_1(p))$ which has no old forms, so $\mathbf{T}^0$-eigenforms are automatically $\mathbf{T}$-eigenforms.

Let f be a Hecke eigenform. A special property of f is that the eigenvalues for T_n can be read off from its Fourier coefficients: $a_n(f)/a_1(f)$. If $a_1(f) = 1$, we say that f is *normalized*. Let f be a normalized cuspidal eigenform. Define $K_f = \mathbf{Q}(\{a_m(f) : m \geq 1\})$ to be the Hecke field of f. As we will see, K_f is a finite extension of $\mathbf{Q}$.

Fact. *The Eisenstein series $G_k(z)$ are Hecke eigenforms.*

It seems no surprise, then, that we will be able to cook up, from $G_k(z)$, a cusp form that is also a Hecke eigenform. The following theorem is the main result of this section, the proof of which is technical, so we omit.

Theorem (3.7 of Ribet)**.** *Suppose that $p \mid B_k$. Then there exists a cusp form $f = \sum_{n \geq 1} a_n q^n$ of weight 2 of some type ε that is a normalized Hecke eigenform which satisfies*

$$a_\ell \equiv 1 + \ell^{k-1} \equiv 1 + \varepsilon(\ell)\ell \mod \mathfrak{p}$$

for all primes $\ell \neq p$, where $\mathfrak{p}$ is some prime in K_f over p that does not depend on ℓ¿.

(3) $\Rightarrow$ (2)

Now we bid farewell to the world of modular forms shortly, and study the number field side of things.

The correspondence between abelian extensions of a field K and the ideal class group of K is given by class field theory. Recall that we hope to prove that if $p \mid B_k$, then $C(\chi^{1-k}) \neq 0$, where C is the p-part of the ideal class group of the cyclotomic field $\mathbf{Q}(\mu_p)$. Therefore, this non-triviality of a certain part of the class group of $\mathbf{Q}(\mu_p)$ can be phrased in terms of the existence of a certain abelian extension of $\mathbf{Q}(\mu_p)$.

Theorem (1.2 of Ribet)**.** *If $p \mid B_k$, then there is a Galois extension $E/\mathbf{Q}$ containing $\mathbf{Q}(\mu_p)$ that satisfies:*

(i) the extension $E/\mathbf{Q}(\mu_p)$ is unramified and $\mathrm{Gal}(E/\mathbf{Q}(\mu_p))$ is a nontrivial abelian group of the form $\mathbf{Z}/p\mathbf{Z} \times \cdots \times \mathbf{Z}/p\mathbf{Z}$, i.e., killed by p;

(ii) if $\sigma \in \mathrm{Gal}(E/\mathbf{Q})$ and $\tau \in \mathrm{Gal}(E/\mathbf{Q}(\mu_p))$, then $\sigma\tau\sigma^{-1} = \chi(\sigma)^{1-k}\tau$.

Using class field theory, one can easily show that this theorem implies that $C(\chi^{1-k}) \neq 0$. Indeed, the first criterion guarantees that $C \neq 0$, using the fact that the Artin map is an isomorphism from the class group of $\mathbf{Q}(\mu_p)$ and the Galois group $\mathrm{Gal}(H_K/K)$, where K is the Hilbert class field of K.

The second condition on the conjugacy behavior prescribes the nontriviality of the χ^{1-k}-eigenspace of C. This follows from the functoriality of the Artin map: cojugation by σ on the Galois group side corresponds to applying σ on the ideal class group side:

$$\sigma \left[\frac{E/\mathbf{Q}(\mu_p)}{\mathfrak{a}} \right] \sigma^{-1} = \left[\frac{E/\mathbf{Q}(\mu_p)}{\sigma(\mathfrak{a})} \right]$$

.

So it suffices to prove this existence of such Galois extension of $\mathbf{Q}(\mu_p)$. This will come from a Galois representation, to which we now turn our attention.

(4) $\Rightarrow$ (3)

How can we go from a Galois representation of an extension of number fields? Well, there is a natural way of doing so, that is, by looking at the kernel of the representation. We say that a Galois representation ρ of $G_{\mathbf{Q}}$ *cuts out* a field $K \subset \overline{\mathbf{Q}}$ if K is the unique field fixed by $\ker \rho$. We also say that ρ is *unramified* at a prime p if the inertia group $I_p \subset \ker \rho$. Note that if ρ is unramified at p, then p does not ramify in the extension $K/\mathbf{Q}$ cut out by ρ, since $\mathrm{Gal}(K/\mathbf{Q}) = G_{\mathbf{Q}}/\ker \rho$

We want to construct a Galois representation that cuts out our desired Galois extension $E/\mathbf{Q}(\mu_p)$. Note that $E/\mathbf{Q}(\mu_p)$ is required to be unramified, while $\mathbf{Q}(\mu_p)/\mathbf{Q}$ is totally ramified. The representation that we construct thus needs to have highly controlled ramification. The following theorem reflects this.

Theorem (1.3 of Ribet)**.** *Let $p \mid B_k$. Then there is a finite field $\mathbf{F} \supset \mathbf{F}_p$ and a continuous representation*

$$\overline{\rho} : G_{\mathbf{Q}} \to \mathrm{GL}_2(\mathbf{F})$$

that satisfies

(i) $\overline{\rho}$ *is unramified at all primes* $\ell \neq p$*;*

(ii) $\overline{\rho}$ *is reducible but not diagonalizable (over* $\mathbf{F}$*) in such a way that* $\overline{\rho}$ *is isomorphism to*

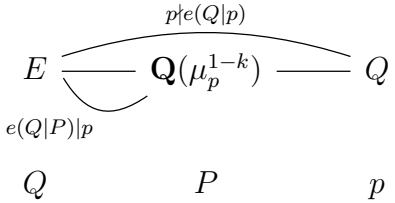
$$\begin{pmatrix} 1 & * \\ 0 & \chi^{k-1} \end{pmatrix};$$

(iii) if D_p *is the decomposition group of* p *in* $G_{\mathbf{Q}}$*, then* $\overline{\rho}|_{D_p}$ *is diagonalizable.*

Note: a variant of Maschke's theorem says that $\overline{\rho}$ is diagonalizable if and only if the image of $\overline{\rho}$ has order prime to $\mathrm{char}(\mathbf{F}) = p$. So (ii) says that the image of $\overline{\rho}$ has order divisible by p, and (iii) says that $\overline{\rho}(D_p)$ has order prime to p.

One can prove that the existence of a Galois representation as such cuts out the desired field extension $E/\mathbf{Q}$. Write $\overline{\rho} : \mathrm{Gal}(E/\mathbf{Q}) \to \mathrm{GL}_2(\mathbf{F})$ for the injection. A rough outline of the argument goes as follows.

- The fixed field of χ^{k-1} (viewed as a character on $\chi_{\mathbf{Q}}$) is $\mathbf{Q}(\mu_p^{1-k})$, so since $\overline{\rho}$ is an extension of χ^{k-1}, we have $\mathbf{Q}(\mu_p^{1-k}) \subset E$.
- Note that $E/\mathbf{Q}$ is finite Galois. Now, if $\sigma \in \mathrm{Gal}(E/\mathbf{Q}(\mu_p^{1-k}))$, then $\overline{\rho}(\sigma)$ becomes a unipotent matrix, and since unipotent matrices form a normal subgroup of $\overline{\rho}(\mathrm{Gal}(E/\mathbf{Q}))$, we see that $E/\mathbf{Q}(\mu_p^{1-k})$ is Galois. Moreover, such unipotent matrices have order p, so $\mathrm{Gal}(E/\mathbf{Q}(\mu_p^{1-k}))$ is killed by p.
- Since the image of $\overline{\rho}$ is divisible by p, and since $\mathrm{Gal}(\mathbf{Q}(\mu_p^{k-1})/\mathbf{Q})$ has order dividing $p-1 = [\mathbf{Q}(\mu_p) : \mathbf{Q}]$, i.e., prime to p, the matrix in the image of $\overline{\rho}$ of order p must come from $\mathrm{Gal}(E/\mathbf{Q}(\mu_p^{1-k}))$. This proves that $E/\mathbf{Q}(\mu_p^{1-k})$ is nontrivial.
- Since $\overline{\rho}$ is unramified outside of p, we only need to show that $E/\mathbf{Q}(\mu_p^{1-k})$ is unramified at p. This comes from (iii), which implies that the order $|D_p| = |\overline{\rho}(D_p)|$ is prime to p. The following scheme shows that the ramification index $e(Q|P) = 1$.



- Finally, the desired conjugation behavior follows from direct computation using (ii).
- Now, we have proved the theorem with $\mathbf{Q}(\mu_p)$ replaced by $\mathbf{Q}(\mu_p^{1-k})$, which is the unique subfield of $\mathbf{Q}(\mu_p)$ of degree $(p-1)/(p-1, k-1)$. To upgrade, we simply consider the compositum $E\mathbf{Q}(\mu_p)$. Then the restriction $\mathrm{Gal}(E\mathbf{Q}(\mu_p)/\mathbf{Q}(\mu_p)) \to \mathrm{Gal}(E/\mathbf{Q}(\mu_p^{1-k}))$ is injective. One can check that everything we need still holds.

⑤ ⇒ ④, the construction

This is really the main crux of the theory, going from Hecke eigenforms to Galois representations. So I hope to spend most of time to explain this. Let us record here a general result on this issue.

Theorem. *Let f be a cuspidal Hecke eigenform of weight $k \geq 1$ and type ε. Let λ be a prime of K_f lying over p and let $\mathcal{O}_\lambda$ denote the completion of $\mathcal{O}_{K_f}$ at λ. There is a unique irreducible 2-dimensional Galois representation*

$$\rho_{f,\lambda} : G_{\mathbf{Q}} \to \mathrm{GL}_2(\mathcal{O}_\lambda)$$

that is unramified at primes $\ell \nmid pN$, and for such an ℓ, if $\mathrm{Frob}_\ell \in G_{\mathbf{Q}}$ is a choice of a Frobenius substitution of ℓ, then the image $\rho_{f,\lambda}(\mathrm{Frob}_\ell)$ satisfies the polynomial equation

$$x^2 - a_\ell(f)x + \epsilon(\ell)\ell^{k-1}.$$

Note that by the Chebotarev density theorem, the knowledge of the behavior of the representation on the Frobenius substitutions prescribes the representation.

This theorem was first proved for weight $k = 2$ by Shimura. The case $k \geq 2$ was conjectured by Serre and proved by Deligne (1971), using étale cohomology. The case $k = 1$ was proved by Deligne and Serre subsequently (1974), building upon the case $k \geq 2$. Since Ribet worked with weight 2 forms, it suffices to explain Shimura's construction of the representation. Even though Ribet's core innovation does not lie in this section, we still explain in detail the ingredients, since this is very seminal.

The key ingredient here is the Eichler-Shimura congruence relation, which describes the reduction of the Hecke action on the Jacobian of a modular curve $X_1(N)$. It roughly states that on the (good) reduction of $X_1(N)$ modulo a prime p, the Hecke correspondence on the Jacobian is given by the Frobenius and its dual. A lot things here need to be unpacked: what is the Hecke action on the Jacobian of modular curve, the details concerning the reduction of an algebraic curve...

Modular curves and their Jacobians

We talked about Hecke actions on the space of modular forms. To formulate Hecke actions on modular curves, we need to understand them in terms of moduli spaces of elliptic curves. Recall that $Y_1(N) = \Gamma_1(N) \setminus \mathcal{H}$ and $X_1(N)$ is the compactified version of $Y_1(N)$. In fact, $Y_1(N)$ is the solution to the moduli problem of enhanced elliptic curves: an *enhanced* elliptic curve for $\Gamma_1(N)$ is a pair (E, Q) where E is a complex elliptic curve and Q is a point of E of order N. We say that two such pairs (E, Q) and (E', Q') are equivalent if there exists an isomorphism $E \to E'$ that maps Q to Q'. The moduli space for the equivalence classes of enhanced elliptic curves for $\Gamma_1(N)$ are in bijection with the modular curve $Y_1(N)$.

With this moduli space interpretation, we can turn $Y_1(N)$ into a Hecke object. The Hecke action T_p on $Y_1(N)$ is a multi-valued function, so it is a correspondence on $Y_1(N)$, i.e., a map

on the divisor groups $T_p : \mathrm{Div}(Y_1(N)) \to \mathrm{Div}(Y_1(N))$ defined as the $\mathbf{Z}$-linear extension of

$$[E, Q] \mapsto \sum_C [E/C, Q + C],$$

where the sum is taken over all subgroups $C \subset E$ of order p such that $C \cap \langle Q \rangle = \{0_E\}$. Likewise, the diamond operator acts on $\mathrm{Div}(Y_1(N))$ via the $\mathbf{Z}$-extension of

$$[E, Q] \mapsto [E, dQ].$$

We also need to turn the Jacobian of $X_1(N)$ into a Hecke object. Recall that the Jacobian of a compact Riemann surface X of genus g is defined to be $\mathrm{Jac}(X) = \Omega^1(X)^\vee / H_1(X, \mathbf{Z})$, which is isomorphic to a g-dimensional complex torus $\mathbf{C}^g/\Lambda_g$. Now consider the compact modular curve $X_1(N)$, since $\Omega^1(X_1(N))$ is isomorphic to $S_2(\Gamma_1(N))$, we have

$$J_1(N) := \mathrm{Jac}(X_1(N)) = S_2(\Gamma_1(N))^\vee / H_1(X_1(N), \mathbf{Z}).$$

Hecke operators act on the dual space $S_2(\Gamma_1(N))^\vee$ by precomposition, and the action factors through $J_1(N)$. The fact that the action factors through $J_1(N)$ has important consequences, for it shows that Hecke algebra consists of endomorphisms of $H_1(X_1(N), \mathbf{Z})$, which is a free finitely generated $\mathbf{Z}$-module:

(i) the (full) Hecke algebra $\mathbf{T_Z} = \mathbf{Z}[\{T_n, \langle n \rangle\}_{n \in \mathbf{Z}}]$ is finitely generated;

(ii) the characteristic polynomial of the T_p's are monic polynomials with integer coefficient, so the eigenvalues of T_p (which, if you recall, appear as coefficients of normalized Hecke eigenforms) are algebraic integers.

For a fixed normalized eigenform f, define an eigenvalue homomorphism $\lambda_f : \mathbf{T_Z} \to \mathbf{C}$ by $Tf = \lambda_f(T) f$. Then we obtain an isomorphism $\mathbf{T_Z}/I_f \to \mathbf{Z}[\{a_n(f)\}_{n \geq 0}]$ of finitely generated $\mathbf{Z}$-modules of rank $[K_f : \mathbf{Q}]$, where I_f is the kernel of λ_f.

Definition. Let f be a normalized eigenform of level N. The *abelian variety associated to* f is the quotient variety

$$A_f = J_1(N)/I_f J_1(N).$$

The action of T_p on $J_1(N)$ becomes $a_p(f)$ when restricted to A_f:

$$\begin{array}{ccc} J_1(N) & \xrightarrow{T_p} & J_1(N) \\ \downarrow & & \downarrow \\ A_f & \xrightarrow{a_p(f)^*} & A_f \end{array},$$

where $(a_p(f)^*\varphi)(g) = a_p(g)^\sigma \varphi(g)$ if $g = f^\sigma$, where $\sigma : K_f \to \mathbf{C}$ is an embedding. In fact, there is a isogenous decomposition

$$J_1(N) \to \bigoplus_f A_f^{m_f},$$

where the sum is taken over a set of representatives $f \in S_2(\Gamma_1(N))$ at levels $M_f \mid N$, and m_f is the number of divisors of N/M_f.

Reduction of algebraic curves and the Eichler-Shimura relation

Another major topic that is involved here is the reduction of curves. The modular curve $X_1(N)$ defined as $\Gamma_1(N) \setminus \mathcal{H}$ is a compact Riemann surface, and thus by the Riemann existence theorem is a complex algebraic curve. In fact, $X_1(N)$ has a model over $\mathbf{Q}$. By a "model over $\mathbf{Q}$" I mean there exists an algebraic curve X defined over $\mathbf{Q}$ (defining equations have $\mathbf{Q}$-coefficients) whose $\mathbf{C}$-points $X(\mathbf{C})$ is isomorphic to $X_1(N)$.

Note: the Hecke operators on $X_1(N)$ are defined over $\mathbf{Q}$ as well. For the Jacobian, we use the Abel-Jacobi theorem $\mathrm{Pic}^0(X) \simeq \mathrm{Jac}(X)$ and the fact that the Picard group can be brought down to $\mathbf{Q}$. From now on, we will be using $\mathrm{Pic}^0(X_1(N))$ in place of $J_1(N)$.

To formulate the Eichler-Shimura congruence relation, we need to reduce the curve $X_1(N)$ over $\mathbf{Q}$ to a curve over finite fields $\mathbf{F}_p$. To do that, we need a model over $\mathbf{Z}$. For elliptic curves, this model is called the global minimal Weierstrass model; for Jacobians or algebraic curves in general, this is called the Néron model or canonical proper model.

Example. The most basic example of an integral model for the modular curve $Y_1(1)$ is given by the scheme $\mathcal{Y} = \mathrm{Spec}(\mathbf{Z}[j])$, and an isomorphism $Y_1(1) \to \mathcal{Y}(\mathbf{C})$ that is given by $\mathrm{SL}_2(\mathbf{Z})z \mapsto (j \mapsto j(z))$.

The theory of canonical models is quite technical and cannot be developed here. For our purpose, it suffices to understand the following. A curve C over $\mathbf{Q}$ defined by equations $\varphi_1, \cdots, \varphi_n$ has *good reduction* at a rational prime p if we can view its $\varphi_1, \cdots, \varphi_n$ as polynomials over $\mathbf{Z}_{(p)}$ (by clearing denominators) so that

(i) the ideal $(\varphi_1, \cdots, \varphi_n)$ is a prime ideal of $\mathbf{Z}_{(p)}[x_1, \cdots, x_m]$,

(ii) and after reducing the coefficients modulo p, the reduced equations $\widetilde{\varphi}_1, \cdots, \widetilde{\varphi}_n$ define a nonsingular algebraic curve over $\mathbf{F}_p$.

We denote the resulting curve over $\mathbf{F}_p$ by $\widetilde{C}$.

Fact. *If C has good reduction at p, then the reduction map $C \to \widetilde{C}$ is surjective, and descends to a surjective map on* $\mathrm{Pic}^0(C) \to \mathrm{Pic}^0(\widetilde{C})$.

For the modular curve $X_1(N)$, the reduction modulo p has the moduli-theoretic interpretation of sending an enhanced elliptic curve (E, Q) to its reduction $(\widetilde{E}, \widetilde{Q})$. If $p \mid N$, then (E, Q) does not reduce well. In fact, we have the following.

Theorem (Igusa)**.** *The modular curve $X_1(N)$ has good reduction at p for $p \nmid N$.*

The Eichler-Shimura congruence relation describes the action of the Hecke operator T_p on the Jacobian of the reduced modular curve.

Theorem (Eichler-Shimura)**.** *Let $p \nmid N$. Let σ_p be the Frobenius map $x \mapsto x^p$ on the coordinates of an algebraic curve over $\mathbf{F}_p$. Then the following diagram commutes:*

$$\begin{array}{ccc} \mathrm{Pic}^0(X_1(N)) & \xrightarrow{T_p} & \mathrm{Pic}^0(X_1(N)) \\ \downarrow & & \downarrow \\ \mathrm{Pic}^0(\widetilde{X_1(N)}) & \xrightarrow{\sigma_{p*}+\widetilde{\langle p\rangle}_*\sigma_p^*} & \mathrm{Pic}^0(\widetilde{X_1(N)}) \end{array}.$$

How to interpret this theorem? Note that T_p is about p-isogenies. In the moduli-theoretic viewpoint, T_p sends $[E, Q]$ to $\sum_C [E/C, Q + C]$, but this is the same as $\sum_{\phi: E \to E'} (E', \phi(Q))$, where the sum is taken over all degree-p isogenies ϕ. Indeed, giving an isogeny of degree p in characteristic 0 is the same as choosing a cyclic subgroup of order p. In characteristic p, an p-isogeny is either the Frobenius, or its dual isogeny (often called the Vershiebung).

One can compute the reduction of $[E/C, Q + C]$ modulo p to see that the reduction $\widetilde{T}_p$, if exists, must be of the form above. But the existence of a well-defined Hecke action reduced over $\mathbf{F}_p$ is a technical issue and involves the study of bad reductions.

The resulting Galois representation

We are finally prepared to construct the desired Galois representation. We know work with level p with $p \mid B_k$, and let f be the cusp eigenform we constructed in (6) $\Rightarrow$ (5).

Definition. The *ℓ-adic Tate module* of the modular curve $X_1(p)$ is

$$V_\ell(\mathrm{Pic}^0(X_1(p))) = \varprojlim_n \mathrm{Pic}^0(X_1(p))[\ell^n] \otimes \mathbf{Q}.$$

Since $\mathrm{Pic}^0(X_1(p))[\ell^n] \simeq (\mathbf{Z}/\ell^n\mathbf{Z})^{2g}$, where g is the genus of $X_1(p)$, so the ℓ-adic Tate module is a free $\mathbf{Q}_\ell$-vector space of dimension $2g$.

The natural action of $G_\mathbf{Q}$ on $\mathrm{Pic}^0(X_1(p))(\overline{\mathbf{Q}})$ induces an automorphism on the ℓ^n-torsion points, and is compatible with the inverse limit defining the Tate module, i.e. the following diagram commutes:

$$\begin{array}{ccc} & G_\mathbf{Q} & \\ \swarrow & & \searrow \\ \mathrm{Aut}(\mathrm{Pic}^0(X_1(p))[\ell^n]) & \longleftarrow & \mathrm{Aut}(\mathrm{Pic}^0(X_1(p))[\ell^{n+1}]) \end{array}.$$

This gives a continuous ℓ-adic representation $\rho_{X_1(p),\ell} : G_\mathbf{Q} \to \mathrm{GL}_{2g}(\mathbf{Z}_\ell) \subset \mathrm{GL}_{2g}(\mathbf{Q}_\ell)$.

Fact. *The reduction map $\mathrm{Pic}^0(X_1(p))[\ell^n] \to \mathrm{Pic}^0(\widetilde{X_1(p)})[\ell^n]$ on the ℓ^n-torsion points is an isomorphism, where the reduction is modulo $q \nmid \ell p$.*

From this fact and the Eichler-Shimura relation, we can derive some important properties of this representation.

(i) In view of the following commutative diagram,

$$\begin{array}{ccc} D_{\mathfrak{q}} & \longrightarrow & \mathrm{Aut}(\mathrm{Pic}^0(X_1(p))[\ell^n]) \\ \downarrow & & \downarrow \simeq \\ G_{\mathbf{F}_{\mathfrak{q}}} & \longrightarrow & \mathrm{Aut}(\mathrm{Pic}^0(\widetilde{X_1(p)})[\ell^n]) \end{array},$$

we see that $\rho_{X_1(p),\ell}$ is unramified at $q \nmid \ell p$. Here, $\mathfrak{q}$ is any prime ideal of $\overline{\mathbf{Z}}$ over q.

(ii) Applying the Eichler-Shimura relation to $\mathrm{Pic}^0(X_1(p))[\ell^n]$, we see that

$$T_q = \mathrm{Frob}_{\mathfrak{q}} + \langle q \rangle q \mathrm{Frob}_{\mathfrak{q}}^{-1}$$

on $\mathrm{Pic}^0(X_1(p))[\ell^n]$, where $\mathrm{Frob}_{\mathfrak{q}}$ is a Frobenius substitution. Therefore, the image $\rho_{X_1(p),\ell}(\mathrm{Frob}_{\mathfrak{q}})$ satisfies the polynomial equation

$$x^2 - T_q x + \langle q \rangle q = 0.$$

Now, we restrict the Galois action on $\mathrm{Pic}^0(X_1(p))$ to the abelian variety A_f associated to f. The quotient map $\mathrm{Pic}^0(X_1(p)) \to A_f$ is $G_{\mathbf{Q}}$-equivariant, and induces a Galois action on the Tate module $V_\ell(A_f)$. The coefficient field K_f also acts, commuting with the Galois action. Since

$$K_f \otimes_{\mathbf{Q}} \mathbf{Q}_\ell \simeq \prod_{\lambda|\ell} K_{f,\lambda},$$

we obtain a decomposition

$$V_\ell(A_f) \simeq \bigoplus_{\lambda|\ell} V_{f,\lambda}, \qquad \dim_{K_{f,\lambda}} V_{f,\lambda} = 2.$$

Thus each factor gives a two-dimensional ℓ-adic Galois representation

$$\rho_{f,\lambda} : G_{\mathbf{Q}} \to \mathrm{GL}_2(K_{f,\lambda}).$$

Since T_q acts as $a_q(f)$ on A_f and $\langle q \rangle$ acts as $\epsilon(q)$, we see that $\rho_{f,\lambda}(\mathrm{Frob}_{\mathfrak{q}})$ satisfies the polynomial equation

$$x^2 - a_q(f)x + \epsilon(q)q = 0,$$

as desired. This is the precisely the statement of the theorem at the beginning of the section for weight $k = 2$. For $k \geq 2$, the idea is similar, but the machinery is considerably heavier.

⑤ ⇒ ④, the verification

The construction of a Galois representation from a cusp eigenform for weight $k = 2$ was known to Ribet. Ribet's innovation lies in the techniques that he used to prove that such a representation is the precisely representation we want, the one with highly controlled ramification. The first thing we notice is that the representation we want is a mod-p representation, where the representation we constructed was p-adic. So we need to discuss reduction of representations.

Let K be a finite extension of a $\mathbf{Q}_p$ with ring of integers $\mathcal{O}$, uniformizer π, and residue field F. Let V be a free K-module of rank 2. A lattice T in V is a free $\mathcal{O}$-module of rank 2 that generates V over K.

Fact. *Let $\rho : G_{\mathbf{Q}} \to \mathrm{GL}(V)$ be a p-adic Galois representation. Then there exists at least one lattice T of V stable that is $G_{\mathbf{Q}}$-stable.*

We call the representation $\overline{\rho} : G_{\mathbf{Q}} \to \mathrm{GL}(T/\pi T)$ the reduction of ρ attached to T. The Brauer-Nesbitt theorem says that the semisimplification of the reduction does not depend on the choice of the lattice.

We want the reduction of the Galois representation to be reducible but not diagonalizable, and that its semisimplification is given by $1 \oplus \chi^{k-1}$. Ribet's main innovation is the following proposition.

(2.1 of Ribet)**.** *Let G be a compact group and let $\rho : G \to \mathrm{GL}(V)$ be a continuous, irreducible representation on a two-dimensional K-vector space. Suppose that, for one (and hence every) G-stable lattice T, the semisimplification of the reduction is*

$$\overline{\rho}_T^{\mathrm{ss}} \simeq \varphi_1 \oplus \varphi_2,$$

where $\varphi_1, \varphi_2 : G \to F^\times$ are characters. Then one can choose a G-stable lattice L whose reduction has the form

$$\overline{\rho}_L(g) = \begin{pmatrix} \varphi_1(g) & b(g) \\ 0 & \varphi_2(g) \end{pmatrix},$$

and is not semisimple.

In other words, even though the semisimplification of a residual representation is independent of the lattice, the extension class is not. The proposition says that we can find a lattice for which the two characters are glued together in a nontrivial way, and, importantly, we can prescribe which character appears on the invariant line. This is sometimes called *Ribet's lemma* or, in Mazur's terminology, Ribet's "wrench."

Proof. Choose any stable lattice T and an $\mathcal{O}$-basis for it. The reduction is reducible since its semisimplification is a sum of two characters. If the reduction has the form

$$\begin{pmatrix} \varphi_1 & * \\ 0 & \varphi_2 \end{pmatrix},$$

we can proceed. Otherwise, we can arrange that it has the form

$$\begin{pmatrix} \varphi_1 & 0 \\ * & \varphi_2 \end{pmatrix}.$$

This second possibility can be changed to the first by changing our lattice. Indeed, putting

$$P = \begin{pmatrix} 1 & 0 \\ 0 & \pi \end{pmatrix}, \qquad P \begin{pmatrix} a & \pi b \\ c & d \end{pmatrix} P^{-1} = \begin{pmatrix} a & b \\ \pi c & d \end{pmatrix},$$

we see that conjugation by P gives an integral representation whose reduction is upper triangular with the required diagonal characters. (Conjugating the matrices simply amounts to choosing another lattice in V.)

Suppose, by contradiction, that *every* such upper-triangular reduction is semisimple. We will construct a sequence of changes of basis which makes the upper-right entries of all the matrices of $\rho(G)$ tend to zero. This will contradict the irreducibility of ρ.

After the first step we may write $\rho(G) \subset \mathrm{GL}_2(\mathcal{O})$ with all lower-left entries divisible by π. Set $M_0 = I$. Inductively, suppose that we have constructed

$$M_n = \begin{pmatrix} 1 & t_n \\ 0 & 1 \end{pmatrix}, \quad t_n \in \mathcal{O},$$

such that, for each $g \in G$,

$$M_n \rho(g) M_n^{-1} = \begin{pmatrix} a_g & \pi^n b_g \\ \pi c_g & d_g \end{pmatrix}, \quad a_g, b_g, c_g, d_g \in \mathcal{O}.$$

Conjugation by P^n changes these matrices into

$$P^n M_n \rho(g) M_n^{-1} P^{-n} = \begin{pmatrix} a_g & b_g \\ \pi^{n+1} c_g & d_g \end{pmatrix}.$$

By our assumption, the reduction of this representation is split. Thus there is a $u_n \in \mathcal{O}$ such that conjugation by

$$U_n = \begin{pmatrix} 1 & u_n \\ 0 & 1 \end{pmatrix}$$

makes the upper-right entries divisible by π; the lower-left entries remain divisible by π^{n+1}. Conjugating back by P^{-n}, we obtain matrices with upper-right entries divisible by π^{n+1} and lower-left entries divisible by π. In other words, the induction works with

$$M_{n+1} = P^{-n} U_n P^n M_n = \begin{pmatrix} 1 & t_n + \pi^n u_n \\ 0 & 1 \end{pmatrix}.$$

Now $t_{n+1} - t_n \in \pi^n \mathcal{O}$, so (t_n) is a Cauchy sequence in the complete ring $\mathcal{O}$. Let $t = \lim_n t_n$ and $M = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$. In the limit, every matrix $M\rho(g)M^{-1}$ has upper-right entry zero. Thus ρ preserves a one-dimensional K-subspace of V, contradicting its irreducibility. □

Applying the lemma to the Galois representation

Now return to the cusp eigenform $f = \sum_{n \geq 1} a_n q^n$ obtained from the Eisenstein congruence. Recall that it is a normalized eigenform of weight 2, level p, and type $\epsilon = \omega^{k-2}$, and that

$$a_\ell(f) \equiv 1 + \ell^{k-1} \pmod{\mathfrak{p}}$$

for each prime $\ell \neq p$, where $\mathfrak{p}$ is a fixed prime of the coefficient field K_f above p. From now on we take

$$K = K_{f,\mathfrak{p}}, \qquad \mathcal{O} = \mathcal{O}_K, \qquad F = \mathcal{O}/\pi\mathcal{O},$$

and write $\rho = \rho_{f,\mathfrak{p}} : G_{\mathbf{Q}} \to \mathrm{GL}_2(K)$ for the p-adic Galois representation associated to f. Let $\widetilde{\chi} : G_{\mathbf{Q}} \to \mathbf{Z}_p^\times$ denote the p-adic cyclotomic character; its reduction modulo p is the character χ defined in the introduction.

To apply Ribet's lemma, we need to check that ρ is irreducible and that the semisimplification of its reduction is $1 \oplus \chi^{k-1}$. The second assertion follows relatively easily from Eichler–Shimura. The first needs a little more work.

(4.1 of Ribet). *The K-representation ρ is irreducible.*

Proof. Suppose that it is reducible. Then its semisimplification is the direct sum of two one-dimensional p-adic representations

$$\rho^{\text{ss}} = \rho_1 \oplus \rho_2, \qquad \rho_i : G_{\mathbf{Q}} \to K^{\times}.$$

The ρ_i are abelian p-adic representations arising as subquotients of the Tate module of the abelian variety A_f. By a theorem of Serre on *locally algebraic* abelian representations, they have the form

$$\rho_i = \widetilde{\chi}^{n_i} \eta_i, \qquad n_i \in \mathbf{Z},$$

where η_i is a character of finite order, unramified outside p. (This is the one substantial input about abelian p-adic representations that we will use without proof.)

For each $\ell \neq p$, the Eichler–Shimura relation and the determinant formula give

$$\begin{aligned} a_\ell(f) &= \eta_1(\text{Frob}_\ell)\ell^{n_1} + \eta_2(\text{Frob}_\ell)\ell^{n_2}, \\ \ell\epsilon(\ell) &= \eta_1(\text{Frob}_\ell)\eta_2(\text{Frob}_\ell)\ell^{n_1+n_2}. \end{aligned}$$

Since ϵ and the η_i have finite order, the determinant identity forces $n_1 + n_2 = 1$. After interchanging the characters, we may assume that $n_1 \geq 1$ and $n_2 \leq 0$. The values of η_i are roots of unity, so in any complex embedding we obtain

$$|a_\ell(f)| \geq \ell^{n_1} - \ell^{n_2} \geq \ell - 1.$$

But f is a cuspidal eigenform of weight 2. The Ramanujan–Petersson bound, proved in this setting by Deligne, says that

$$|a_\ell(f)| \leq 2\sqrt{\ell} \qquad (\ell \neq p).$$

For every prime $\ell \geq 7$ we have $\ell - 1 > 2\sqrt{\ell}$, a contradiction. Hence ρ is irreducible. □

Let $T \subset V$ be any $G_{\mathbf{Q}}$-stable lattice. For every prime $\ell \neq p$, we know from our construction that

$$\begin{aligned} \operatorname{tr} \overline{\rho}_T(\text{Frob}_\ell) &\equiv a_\ell(f) \equiv 1 + \ell^{k-1} \pmod{\pi}, \\ \det \overline{\rho}_T(\text{Frob}_\ell) &\equiv \ell\epsilon(\ell) \equiv \ell^{k-1} \pmod{\pi}. \end{aligned}$$

Here we used $\epsilon(\ell) = \omega^{k-2}(\ell)$, which reduces to ℓ^{k-2} modulo $\mathfrak{p}$. Since $\chi(\text{Frob}_\ell) = \ell \pmod{p}$, the characteristic polynomial is

$$\det(x - \overline{\rho}_T(\text{Frob}_\ell)) = (x-1)(x - \chi^{k-1}(\text{Frob}_\ell)).$$

By the Chebotarev density theorem, the Frobenius elements at primes away from p form a dense collection of conjugacy classes in $G_{\mathbf{Q}}$. It follows that the characteristic polynomials of $\overline{\rho}_T(g)$ and of $(1 \oplus \chi^{k-1})(g)$ agree for every $g \in G_{\mathbf{Q}}$. Brauer–Nesbitt therefore gives

$$\overline{\rho}_T^{\mathrm{ss}} \simeq 1 \oplus \chi^{k-1}.$$

In particular, every reduction has two one-dimensional Jordan–Hölder factors, but it need not be a direct sum. Combining this with the irreducibility proposition and Ribet's lemma, we can make the following crucial choice.

(4.2 of Ribet)**.** *There exists a $G_{\mathbf{Q}}$-stable $\mathcal{O}$-lattice $L \subset V$ such that, with $M = L/\pi L$, the action of $G_{\mathbf{Q}}$ on M is given, in a suitable basis, by*

$$\boxed{\overline{\rho}(g) = \begin{pmatrix} 1 & b(g) \\ 0 & \chi^{k-1}(g) \end{pmatrix}, \quad g \in G_{\mathbf{Q}},}$$

and this representation is not semisimple.

What exactly does the last condition mean? If the upper-right entry could be eliminated by one fixed change of basis, then the reduction would become $1 \oplus \chi^{k-1}$. This is impossible for our chosen lattice. In fact, the matrices above satisfy

$$b(gh) = b(h) + b(g)\chi^{k-1}(h),$$

so the function b contains the information of an extension of two Galois characters. The semisimplification remembers only the diagonal entries; the nonzero extension class is precisely the additional arithmetic information that Ribet has extracted by changing lattices.

At this point we have nearly obtained the representation required in (4). It is unramified away from p, because the original Tate-module representation is, and it is reducible but not semisimple. What remains is a subtle condition at the prime p: *the restriction of $\overline{\rho}$ to a decomposition group at p must be semisimple.* Global non-splitting, by itself, says nothing about this local condition.

The local condition at p: finite flat group schemes

We now explain the final argument of Ribet's paper, which uses the geometry of the modular Jacobian at p. This is the part of the proof where the good-reduction theory of Deligne–Rapoport and the classification theorem of Raynaud enter. The complete proofs of these geometric theorems are beyond our scope, but we can explain the exact role they play.

Choose a prime of $\overline{\mathbf{Q}}$ above p, and let $D_p \subset G_{\mathbf{Q}}$ be its decomposition group. Write

$$K^+ = \mathbf{Q}(\mu_p)^+$$

for the maximal real subfield of $\mathbf{Q}(\mu_p)$, and put

$$D = D_p \cap \mathrm{Gal}(\overline{\mathbf{Q}}/K^+).$$

The prime p is totally ramified in $K^+/\mathbf{Q}$, and

$$[D_p : D] = [K^+ : \mathbf{Q}] = \frac{p-1}{2}.$$

In particular, this index is prime to p. Consequently, it is enough to show that the image of D in $\mathrm{GL}(M)$ has order prime to p: the same will then be true of the image of D_p.

Let E be the completion of K^+ at its unique prime above p. Then

$$D \simeq \mathrm{Gal}(\overline{E}/E), \qquad e(E/\mathbf{Q}_p) = \frac{p-1}{2} < p-1.$$

The inequality for the ramification index is the reason for working over the *real* cyclotomic field rather than the full cyclotomic field. It is exactly the hypothesis needed in Raynaud's theorem below.

Recall that A_f is the abelian variety associated to the eigenform f, constructed from the Jacobian of $X_1(p)$. At the prime p the modular curve has bad reduction, so the simple good-reduction argument from our discussion of Eichler–Shimura cannot be applied directly. However, Deligne and Rapoport proved a much more precise fact about the geometry at this prime.

Theorem (Deligne–Rapoport)**.** *The abelian variety A_f acquires good reduction over $K^+ = \mathbf{Q}(\mu_p)^+$. In particular, after extending scalars to E, it is the generic fiber of an abelian scheme $\mathcal{A}$ over the ring of integers $\mathcal{O}_E$.*

Let me recall what this means. An abelian scheme $\mathcal{A}$ over $\mathcal{O}_E$ is a smooth and proper group scheme whose generic fiber is the abelian variety A_f/E and whose special fiber is again an abelian variety. For an integer $n \geq 1$, multiplication by n on $\mathcal{A}$ has a scheme-theoretic kernel

$$\mathcal{A}[n] = \ker([n] : \mathcal{A} \to \mathcal{A}).$$

It is a finite flat group scheme over $\mathcal{O}_E$. Its geometric generic fiber is the usual finite Galois module $A_f[n](\overline{E})$. The word *flat* is important: it is what allows a group of torsion points in characteristic zero to be studied together with its reduction in characteristic p.

We need a finite flat group scheme whose generic fiber is not necessarily all of $A_f[p]$, but rather the particular two-dimensional residual representation $M = L/\pi L$ obtained from Ribet's lemma. One might worry that our choice of lattice L was made purely by linear algebra and has no geometric meaning. The following proposition resolves this issue.

(4.3 of Ribet)**.** *The D-module M is the Galois module attached to a finite flat commutative group scheme $\mathcal{M}$ over $\mathcal{O}_E$, killed by p, with an action of the finite field F. In particular,*

$$\mathcal{M}(\overline{E}) \simeq M$$

as F-vector spaces with D-action.

Explanation of the construction. The coefficient field K_f acts on A_f up to isogeny: we have an embedding

$$K_f \hookrightarrow \mathrm{End}_{\mathbf{Q}}(A_f) \otimes_{\mathbf{Z}} \mathbf{Q}.$$

After replacing A_f by a $\mathbf{Q}$-isogenous abelian variety, we may arrange that its endomorphism ring contains the full ring of integers of K_f. Moreover, $G_{\mathbf{Q}}$-stable lattices in the p-adic Tate module correspond, up to scaling, to choices of isogenous abelian varieties with $G_{\mathbf{Q}}$-stable finite kernels. Thus we may make a further isogeny so that our particular lattice L is realized by the $\mathfrak{p}$-component of the Tate module. In this realization, $M = L/\pi L$ is the $\mathfrak{p}$-torsion Galois module, and hence is a subgroup of the p-power torsion of A_f. Since M is killed by π, it is also killed by p.

The property of having good reduction over E is preserved under isogeny, so this new abelian variety also extends to an abelian scheme $\mathcal{A}/\mathcal{O}_E$. We have a finite flat group scheme $\mathcal{A}[p]$, and we take the *scheme-theoretic closure* of the generic-fiber subgroup M inside $\mathcal{A}[p]$. Denote this closure by $\mathcal{M}$.

Why is $\mathcal{M}$ flat? Here is a useful elementary observation. If R is a discrete valuation ring with fraction field K, and a closed subscheme of $\mathrm{Spec}(B \otimes_R K)$ is given by an ideal I, its scheme-theoretic closure in $\mathrm{Spec}(B)$ is defined by $I \cap B$. The quotient $B/(I \cap B)$ has no R-torsion: if $ra \in I \cap B$ for some nonzero $r \in R$, then $a \in I \cap B$. Over a discrete valuation ring, torsion-free modules are flat. Thus the closure is flat. In our situation it is also finite, since it is closed inside the finite scheme $\mathcal{A}[p]$, and it is a group scheme because the generic-fiber subgroup is stable under the group operations.

Finally, the action of the integer ring of K_f extends to $\mathcal{A}$ and preserves $\mathcal{M}$. On M the ideal $\mathfrak{p}$ acts by zero, so the action factors through its residue field F. We therefore obtain the required finite flat F-module scheme. □

The benefit of working with a group *scheme*, rather than just the group of its geometric points, is that we can distinguish the connected part from the *étale* part of its reduction. A finite flat group scheme $\mathcal{G}$ over a complete discrete valuation ring has a canonical connected–étale exact sequence

$$0 \longrightarrow \mathcal{G}^0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{G}^{\text{ét}} \longrightarrow 0.$$

Here $\mathcal{G}^0$ is the identity component (determined by the special fiber), and $\mathcal{G}^{\text{ét}}$ is the maximal finite étale quotient. Notice that this decomposition is a statement about the *integral model*: in characteristic zero all of these finite group schemes have étale geometric generic fibers. If $\mathcal{G}$ has an F-action, then the connected–étale sequence is compatible with this action.

We will use two facts about finite flat group schemes:

(i) If $\mathcal{G}$ is finite étale over $\mathcal{O}_E$, then $\mathcal{G}(\overline{E})$ is an *unramified* representation of D. Indeed, a finite étale cover of $\mathrm{Spec}(\mathcal{O}_E)$ corresponds to a finite unramified extension of E.

(ii) *Raynaud's theorem:* if $e(E/\mathbf{Q}_p) < p - 1$, a finite commutative group scheme over E killed by p has at most one finite flat model over $\mathcal{O}_E$. Equivalently, in this range the generic fiber determines a finite flat model uniquely (up to isomorphism).

The second statement is quite remarkable. In general, the generic fiber of a finite flat group scheme does not uniquely determine its reduction. Raynaud tells us that when the ramification is sufficiently small, it does. This is the key to the last step.

Theorem (4.4 of Ribet)**.** *The image of D in $\mathrm{Aut}_F(M)$ has order prime to p. Consequently the restriction $\overline{\rho}|_{D_p}$ is semisimple.*

Proof. Recall that M has a one-dimensional D-stable subspace X on which D acts trivially, and that the quotient $Y = M/X$ carries the character χ^{k-1}. Thus

$$0 \longrightarrow X \simeq F \longrightarrow M \longrightarrow Y \simeq F(\chi^{k-1}) \longrightarrow 0.$$

Let $\mathcal{X}$ be the scheme-theoretic closure of X in $\mathcal{M}$. By the argument above, $\mathcal{X}$ is a finite flat subgroup scheme. Its geometric generic fiber is a *trivial* D-module, so it has an obvious finite flat model over $\mathcal{O}_E$: the constant group scheme attached to the additive group of F. Since $e(E/\mathbf{Q}_p) < p-1$, Raynaud's theorem says that this model is unique. Consequently, $\mathcal{X}$ is itself constant and in particular is nontrivial and étale.

This shows that $\mathcal{M}$ cannot be connected. Indeed, the special fiber of a connected finite group scheme cannot contain a nontrivial finite étale subgroup. Consider now its connected–étale exact sequence

$$0 \longrightarrow \mathcal{M}^0 \longrightarrow \mathcal{M} \longrightarrow \mathcal{M}^{\text{ét}} \longrightarrow 0.$$

Passing to geometric generic fibers, we get an exact sequence of D-modules and F-vector spaces

$$0 \longrightarrow M^0 \longrightarrow M \longrightarrow M^{\text{ét}} \longrightarrow 0.$$

Because $\mathcal{M}$ is not connected, M^0 is a proper subspace of M. On the other hand, M^0 cannot be zero. Otherwise $\mathcal{M}$ would be finite étale, so M would be unramified, whereas its quotient Y is ramified over E.

Let us verify this last point. The extension $\mathbf{Q}_p(\mu_p)/E$ is a totally ramified quadratic extension: E is the completion of the maximal real subfield. Hence the image of D under the mod-p cyclotomic character χ is $\{\pm 1\}$, with the nontrivial value occurring already on inertia. Since k is even, $k-1$ is odd, and therefore $\chi^{k-1}|_D$ is nontrivial on inertia. So Y is indeed ramified.

We have shown that M^0 is a nonzero proper F-subspace of the two-dimensional space M. Hence

$$\dim_F M^0 = \dim_F M^{\text{ét}} = 1.$$

Moreover, M^0 cannot equal X. If it did, $M/X \simeq M^{\text{ét}}$ would be unramified, contrary to the ramification of $Y = M/X$. We have therefore produced *two distinct* D-stable lines in M: the original line X, and the new line M^0 coming from the connected component of $\mathcal{M}$.

Two distinct lines span the two-dimensional vector space M. Thus D acts diagonally with respect to this decomposition. The characters on the two lines are necessarily 1 and χ^{k-1}, so

$$\overline{\rho}|_D \simeq 1 \oplus \chi^{k-1}|_D.$$

In particular, its image has order prime to p (in fact, it has order at most 2). Since $[D_p : D] = (p-1)/2$ is also prime to p, the image of D_p has order prime to p. By Maschke's theorem, $\overline{\rho}|_{D_p}$ is semisimple, as required. $\square$

It is useful to point out the contrast here. The same residual representation M is *globally* a non-split extension of χ^{k-1} by the trivial character, but it *locally* splits at p. The tension between these two facts is exactly what will produce an unramified extension of the cyclotomic field.

(4) $\Rightarrow$ (3) $\Rightarrow$ (2), the conclusion

We now finish the argument using the class field theory discussed earlier. We have constructed a continuous representation

$$\overline{\rho} : G_{\mathbf{Q}} \longrightarrow \mathrm{GL}_2(F), \qquad \overline{\rho}(g) = \begin{pmatrix} 1 & b(g) \\ 0 & \chi^{k-1}(g) \end{pmatrix}$$

with the following properties:

(i) $\overline{\rho}$ is unramified at every rational prime $\ell \neq p$;

(ii) $\overline{\rho}$ is not semisimple;

(iii) the image of a decomposition group at p has order prime to p.

Let $K_0 = \mathbf{Q}(\mu_p)$ and $G_{K_0} = \mathrm{Gal}(\overline{\mathbf{Q}}/K_0)$. Since $\chi|_{G_{K_0}} = 1$, the restriction of our representation has the particularly simple form

$$\overline{\rho}(h) = \begin{pmatrix} 1 & b(h) \\ 0 & 1 \end{pmatrix}, \qquad h \in G_{K_0}.$$

The product of two such matrices amounts to addition of their upper-right entries. Hence

$$b|_{G_{K_0}} : G_{K_0} \longrightarrow (F, +)$$

is a continuous group homomorphism. It is *nonzero*. For if its image were zero, $\overline{\rho}$ would factor through

$$G_{\mathbf{Q}}/G_{K_0} = \mathrm{Gal}(K_0/\mathbf{Q}),$$

a group of order $p-1$. By Maschke's theorem this representation would be semisimple, contrary to the choice of lattice.

Let H be the kernel of $b|_{G_{K_0}}$ and let $L = \overline{\mathbf{Q}}^{H}$ be its fixed field. Since $H = G_{K_0} \cap \ker \overline{\rho}$, it is normal in $G_{\mathbf{Q}}$. Therefore we have a tower of Galois extensions

$$\begin{array}{c} L \\ | \\ K_0 = \mathbf{Q}(\mu_p) \\ | \\ \mathbf{Q} \end{array}$$

with $L/\mathbf{Q}$ Galois and

$$\mathrm{Gal}(L/K_0) \simeq b(G_{K_0}) \subset (F,+).$$

In particular, L/K_0 is a nontrivial abelian extension and every element of its Galois group is killed by p.

Why is this extension unramified? For primes above $\ell \neq p$, the representation $\overline{\rho}$ is unramified, so L/K_0 is unramified there. At the prime above p, an inertia group for L/K_0 maps under $\overline{\rho}$ into the group of upper unipotent matrices. Every nonidentity matrix of this kind has order p. On the other hand, we just proved that the image of the entire decomposition group D_p has order prime to p. The inertia image must therefore be trivial. This proves that L/K_0 *is everywhere unramified*.

Finally, we must verify that the action of $\mathrm{Gal}(K_0/\mathbf{Q})$ on $\mathrm{Gal}(L/K_0)$ is the correct one. If

$$A = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}, \qquad U(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

then a direct matrix computation gives

$$AU(x)A^{-1} = U(ad^{-1}x).$$

Apply this with $A = \overline{\rho}(g)$, where $g \in G_{\mathbf{Q}}$, and $U(x) = \overline{\rho}(h)$, where $h \in G_{K_0}$. Since $a = 1$ and $d = \chi^{k-1}(g)$, we obtain

$$b(ghg^{-1}) = \chi^{1-k}(g)b(h).$$

Thus the conjugation action of $\mathrm{Gal}(K_0/\mathbf{Q})$ on $\mathrm{Gal}(L/K_0)$ is given precisely by the character χ^{1-k}.

By class field theory, the Galois group of any abelian unramified extension of K_0 is a quotient of its ideal class group A. Since $\mathrm{Gal}(L/K_0)$ is killed by p, the reciprocity map factors through the quotient $C = A/A^p$, giving a surjective, $\mathrm{Gal}(K_0/\mathbf{Q})$-equivariant map

$$C \twoheadrightarrow \mathrm{Gal}(L/K_0).$$

But the right-hand side is nonzero and is entirely in the χ^{1-k}-isotypical component. Since $\mathrm{Gal}(K_0/\mathbf{Q})$ has order prime to p, the decomposition into isotypical components is exact. Hence

$$\boxed{C(\chi^{1-k}) \neq 0.}$$

This is precisely the converse to Herbrand's theorem that we wanted to prove. □

Let me end by emphasizing what has happened. A divisibility property of the Bernoulli number B_k first produces a congruence between an Eisenstein series and a cuspidal eigenform. The geometry of modular curves turns this eigenform into a p-adic Galois representation. Ribet's lemma then extracts from it a nontrivial residual extension of two characters, and Raynaud's theorem shows that this extension becomes split locally at p. The resulting Galois representation cuts out a nontrivial unramified abelian p-extension of $\mathbf{Q}(\mu_p)$, which class field theory identifies with a quotient of the appropriate eigenspace of the class group. This completes Mazur's hexagon: we have come back from modular forms to the arithmetic of cyclotomic fields.